et al., 2024), we evaluate whether SAEs can learn monosemantic features by using datasets with polysemous words – where the same token has different meanings – such as WiC.

3 PRELIMINARIES

We here summarize the key concepts and notation required to understand SAE architectures and their training procedures. The central goal of SAEs during training is to decompose a model's activation $\mathbf{x} \in \mathbb{R}^n$ into a sparse, linear combination of feature directions, expressed as:

$$\mathbf{x} \approx \mathbf{x_0} + \sum_{i=1}^{M} f_i(\mathbf{x})\mathbf{d}_i, \tag{1}$$

where $\mathbf{d}_i$ represents unit-norm latent feature directions ($M \gg n$), and the sparse coefficients $f_i(\mathbf{x}) \geq 0$ correspond to feature activations for the input $\mathbf{x}$. This framework aligns with the structure of an autoencoder, where the input $\mathbf{x}$ is encoded into a sparse vector $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^M$ and decoded to approximate the original input.

3.1 ARCHITECTURE

The typical SAE is structured as a single-layer autoencoder, where the encoding and decoding functions are defined as follows:

$$\mathbf{f}(\mathbf{x}) := \sigma(W_{\text{enc}}(\mathbf{x} - \mathbf{b}_{\text{dec}}) + \mathbf{b}_{\text{enc}}), \tag{2}$$

$$\hat{\mathbf{x}}(\mathbf{f}(\mathbf{x})) := W_{\text{dec}}\mathbf{f}(\mathbf{x}) + \mathbf{b}_{\text{dec}}. \tag{3}$$

Here, $W_{\text{enc}} \in \mathbb{R}^{Rn \times n}$ and $W_{\text{dec}} \in \mathbb{R}^{n \times Rn}$ are the encoding and decoding weight matrices, respectively, where R is the expand ratio that determines the dimension of the intermediate in SAE. The vectors $\mathbf{b}_{\text{enc}}$ and $\mathbf{b}_{\text{dec}}$ represent the biases. We initialize the W_{dec} as the transposed W_{enc}, following the methodology from previous studies (Gao et al., 2024). The sparse encoding $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^{Rn}$ serves as the intermediate representation, while $\hat{\mathbf{x}}(\mathbf{f})$ reconstructs the input $\mathbf{x}$ from the sparse encoding. The columns of W_{dec} correspond to the learned feature directions $\mathbf{d}_i$, and the elements of $\mathbf{f}(\mathbf{x})$ represent the activations for these directions. $\sigma(\cdot)$ is an activation function; we adopt ReLU, TopK (Gao et al., 2024), or JumpReLU, including its STE variant (Rajamanoharan et al., 2024b) in this paper, while using ReLU by default.

3.2 TRAINING METHODOLOGY

Training SAE involves optimizing a loss function that balances reconstruction accuracy and sparsity. The reconstruction loss is defined as the squared error between the input $\mathbf{x}$ and the reconstruction $\hat{\mathbf{x}}(\mathbf{f}(\mathbf{x}))$ and sparsity in the activations $\mathbf{f}(\mathbf{x})$ is encouraged using the L_1 norm. The overall loss function is a combination of these two terms, weighted by a sparsity regularization factor λ:

$$\mathcal{L}_{\text{SAE}}(\mathbf{x}) = \mathcal{L}_{\text{recon}}(\mathbf{x}) + \lambda\mathcal{L}_{\text{sparse}}(\mathbf{f}(\mathbf{x})) = \|\mathbf{x} - \hat{\mathbf{x}}(\mathbf{f}(\mathbf{x}))\|_2^2 + \lambda\|\mathbf{f}(\mathbf{x})\|_1. \tag{4}$$

This formulation encourages the autoencoder to reconstruct the input faithfully while enforcing sparsity in the latent features. To prevent trivial solutions, such as reducing the sparsity penalty by scaling down the encoder outputs and increasing the decoder weights, it is necessary to constrain the norms of the columns of W_{dec} during training. In addition, we incorporate an auxiliary loss known as Ghost Grads (Jermyn & Templeton, 2024), which reduces the occurrence of dead latents where SAE features stop activating for long periods. In Appendix A, we discusses the relationship between dead latents and Ghost Grads.

Following prior work (Templeton et al., 2024), we use an expand ratio of $R = 32$ and a sparsity regularization factor of $\lambda = 0.05$ by default for training SAE. The base LLM used as activations for the SAE is GPT-2 small (Radford et al., 2019). Unless specified otherwise, activations are extracted from the 4th layer. Inspired by prior work (Makelov et al., 2024), the training data consisted of the WiC dataset (Raganato et al., 2020) for in-domain data (i.e., data from the same domain as the evaluation dataset) and RedPajama (Weber et al., 2024) for open-domain data (i.e., data from a general domain used during the pre-training of LLMs). The distinction between open-domain and in-domain data is further explained in Appendix B. By default, we use in-domain data for training. Additional training details, such as hyper-parameters, are provided in Appendix C.

Table 1: Statistics and examples of `poly-contexts` and `mono-contexts` in PS-Eval.

# of Total Samples	1112 (label 0: 556, label 1: 556)
Example Sentences	`poly-contexts` (**space**) Context 1: *The astronauts walked in outer "space" without a tether.* Context 2: *The "space" between his teeth.* Label: 0 (**Different**)
	`mono-contexts` (**space**) Context 1: *They stopped at an open "space" in the jungle.* Context 2: *The "space" between his teeth.* Label: 1 (**Same**)

3.3 Trade-off between Reconstruction and Sparsity

As shown in Equation 4, a critical aspect of evaluating SAE during training lies in balancing two primary metrics: mean squared error for reconstruction accuracy and the L_0 norm for measuring sparsity. These metrics often exhibit a trade-off achieving lower MSE typically results in less sparsity, and increasing sparsity generally raises the reconstruction error. This trade-off can be represented by the Pareto frontier, as shown in Figure 2, where the x-axis corresponds to the L_0 norm (sparsity) and the y-axis represents MSE (reconstruction error). Ideally, optimal models occupy the lower-left part of the figure, where both MSE and sparsity are minimized simultaneously.

To improve the Pareto frontier, there are two main strategies: increasing the number of latent variables and enhancing activation function. For increasing latent variables, scaling laws have shown that higher dimensionality improve this Pareto frontier.

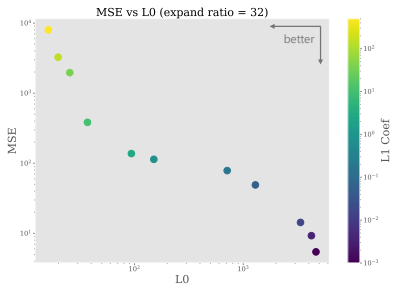

Figure 2: MSE v.s. L_0 for expand ratio = 32 across varying L_1 coefficients (λ). The color gradient represents the L_1 coefficient values, with brighter colors indicating larger λ. As the λ increases, the model becomes sparser (lower L_0) while MSE gets worse.

4 PS-Eval

As a holistic evaluation of SAE considering the semantics in the features, we introduce Poly-Semantic Evaluation (PS-Eval), a suite of datasets and metrics for evaluation. PS-Eval consists of polysemous words whose meanings are different based on the given context, which aims to evaluate whether SAE can map entangled polysemantic neurons in LLMs into simple monosemantic features for better interpretability while distinguishing the different meanings.

4.1 Dataset

We build PS-Eval on top of the WiC dataset (Pilehvar & Camacho-Collados, 2019), which is a rich source of polysemous words. The original WiC dataset provides two contexts with a shared target word for each sample, and its task is to determine whether the target word holds the same meaning or a different meaning in both contexts. The dataset has binary labels, where a label of **0** indicates that the target word has **different** meanings in the two contexts, and a label of **1** signifies that the target word's meaning remains the **same** in both context. For PS-Eval, we carefully select instances from the WiC dataset whose target word is tokenized as a single token in GPT-2 small.

We refer to the two contexts with a label of 0 (indicating different meanings) as `poly-contexts` (**target word**). Table 1 shows an example using the target word "space" in poly-context. Even though the tokenization of "space" remains the same in both contexts, the word carries different meanings: in the first context, it refers to "universe" while in the second, it refers to "an area or gap". On the other hand, when the label is 1, the meaning of the target word remains the same across both contexts. We denote these as `mono-contexts` (**target word**). Table 1 also has

Table 2: (**Left**) Confusion matrix for polysemy detection in SAE evaluation. Note that *Positive* or *Negative* labels are just for convenience to describe the confusion matrix, and these do not mean that, it is bad to identify that the different words have different features or it is good to identify that the same words have the same features. (**Right**) metrics (accuracy, recall, precision, and specificity) and their interpretations.

	Mono-context	**Poly-context**
Same Max Activated Feature	Same meaning, same feature (**True Positive**)	Different meaning, same feature (**False Positive**)
Different Max Activated Feature	Same meaning, different feature (**False Negative**)	Different meaning, different feature (**True Negative**)

Metric	**Formula**	**Interpretation**
Accuracy	$\frac{TP+TN}{TP+FP+TN+FN}$	How well the model predicts the overall situation
Recall	$\frac{TP}{TP+FN}$	When the same meaning's words are given, how often they are identified as the same
Precision	$\frac{TP}{TP+FP}$	When the same features are activated, how often the meanings are same
F1	$\frac{2\times \text{Recall}\times \text{Precision}}{\text{Recall}+\text{Precision}}$	How Recall and Precision are balanced
Specificity	$\frac{TN}{TN+FP}$	When the different meaning's words are given, how often they are identified as different

an example using the target word "space" in mono-contexts. Here, the word "space" refers to "an area or gap" in both contexts, and thus the label is 1, signifying that the meaning is consistent.

4.2 Evaluation Metrics

PS-Eval focuses on analyzing activations of the SAE for the target word across different contexts to determine whether SAE can extract monosemantic features.To identify the activation related to polysemous target words, we consider the following formatted input:

`{context}`. *The* `{target word}` *means*

where the `{context}` is an example sentence of `{target word}`, reflecting either polysemous contexts (poly-context) or monosemous contexts (mono-context). As done in Wang et al. (2022) such a format sentence makes it easy to identify the corresponding activation. In this paper, the activation of the token for the `{target word}` at layer l is denoted as $a^l_{\text{LLM}}(\text{context})$, and the feature from the SAE is represented as $f^l_{\text{SAE}}(\text{context})$. Figure 1 (left) describe the conceptual workflow.

Confusion Matrix for Polysemy Detection To evaluate the ability of SAEs to extract monosemantic features, we construct a confusion matrix (Table 2; left) based on the maximum activations of the target word in different contexts. Each cell in the matrix is defined as follows:

- In mono-context, where the target word has the same meaning in both contexts, we expect the same feature with the highest activation to appear. This corresponds to a True Positive (TP).
- In mono-context, if different features with the highest activation appear, this results in a False Negative (FN).
- In poly-context, where the target word has different meanings, we expect different features with the highest activation to appear, representing a True Negative (TN).
- In poly-context, if the same feature with the highest activation appears for the target word, this results in a False Positive (FP).

Note that *positive* or *negative* labels here are just for convenience to describe the confusion matrix, and these do not mean that, it is undesirable to identify that the different words have different features or it is desirable to identify that the same words have the same features. After classifying each instance into a confusion matrix, we then calculate the quantitative metrics such as **accuracy**, **precision**, **recall**, **F1 score** and **specificity** to assess whether the SAE is capable of acquiring monosemantic features (Table 2; right); Recall measures how often the same meaning words have the same SAE features. Precision measures how often the same SAE features are attributed to the same meaning words. Accuracy and F1 score can quantify the overall performance. Specificity measures how often the **different** meaning words have the **different** SAE features.

If SAEs have ideal features, all of these metrics can be high simultaneously. We advocate for a comprehensive evaluation that considers all metrics holistically, in complement to existing performance measures, such as MSE, L_0, or IOI performances. While measuring individual metrics may also be useful to improve SAEs, we note that we need to carefully treat Specificity when solely interpreting it. Because Specificity measures when the different meaning's words are given, how often they are identified as different, it may reach an unfairly high value by definition if SAE features have significantly different representations of each other. For comparison, we investigate the performance of an SAE that activates *randomly* (see Appendix D) and a *dense* SAE that is trained without L_1